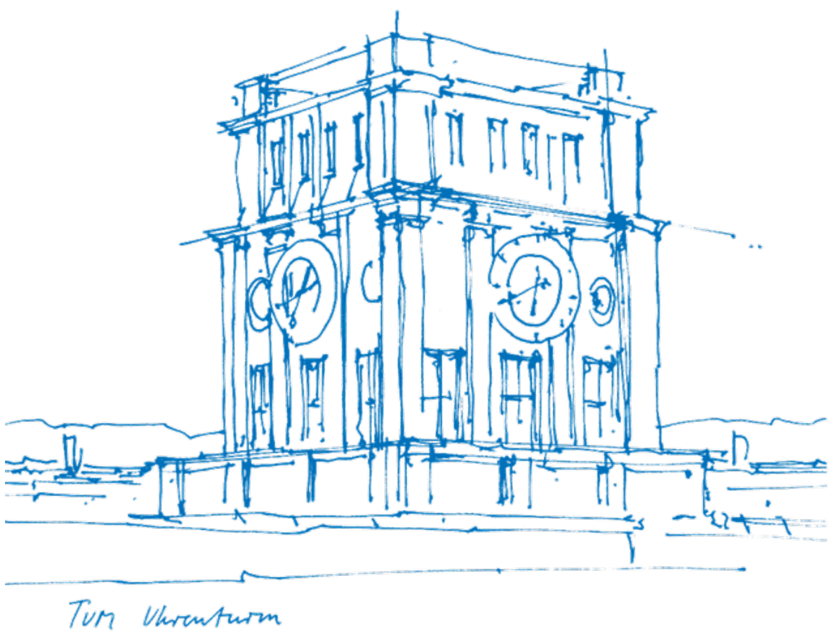
Chair of Information Systems and Business Process Management (i17)
Department of Computer Science
TUM School of Computation, Information and Technology
Technical University of Munich

Master's Thesis in Information Systems

Xin Wen

Document-Based Process Modeler



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Dokumentatenbasierte Bearbeitung und Änderung von Prozessmodellen

Thesis for the Attainment of the Degree
Master of Science

at the TUM School of Computation, Information and Technology,
Department of Computer Science,
Chair of Information Systems and Business Process Management (i17)

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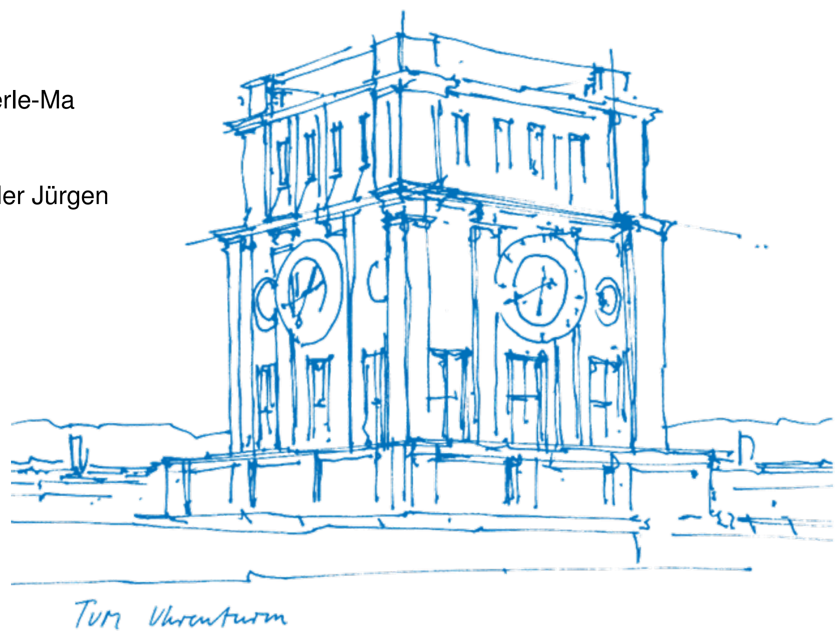

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Abstract

Business process is valuable asset of an organization.But most of them are store in unstructured format, e.g., Word documents, PDF files. Manual translation from unstructured documents to structured process models is time-consuming and error-prone. LLM has shown as a great technique to support creating proces modeling efficiently, but it still cannot fully replace human in process modeling. Hybrid human-LLM process modeling is a promising way to leverage the strength of both human and LLM, and mitigate their weaknesses. However, existing tools and approaches still have limitations in terms of usability, integration, and functionality, especially in the use case for creating process models from unstructured documents. This work aims to address these limitations by proposing and developing an integrated system that supports process modeling from documents using LLMs with a more seamless, efficient and user-friendly workflow.

Keywords: *Process Modeling*

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Introduction

Motivation

Business process is valuable asset of an organization. But most of them are store in unstructured format, e.g., Word documents, PDF files. Munal translate from unstructured documents to structured process models is time-consuming and error-prone. In the era of Gen-AI, LLM has shown as a promising solution to support creating proces modeling efficiently. However, it still cannot fully replace human in process modeling. Hybrid human-LLM process modeling is a promising to leverage the strength of both human and LLM, and mitigate their weaknesses.

However, existing tools and approaches still have limitations in terms of usability, integration, and functionality, especially in the use case for creating process models from documents. For example, user need to manually extract relevant information from documents and input it into the conversational chatbot. Besides, the separate interface of the document and the generated model make it hard to verify the result.

To address these limitations, this work proposes an integrated tool that supports process modeling from documents with a more seamless, efficient, and user-friendly workflow by offering the following core features: (1) allowing users to upload documents and select passages from the documents to serve as input for LLM-based process model generation directly, (2) providing a clear visual link between the text and the generated model through display document and model side by side, and (3) allowing users to establish process-subprocess relationships between models across the documents.

Research Questions

This work addresses the following research questions:

RQ1 How do existing approaches and tools support process modeling from documents?

This question investigates how existing approaches and tools support process modeling from documents by identifying their functionalities and capabilities. It helps guide the design of the proposed system.

RQ2 How can an integrated system be designed to support process modeling from documents?

This question addresses the design of the proposed system. It focuses on what functionalities should be integrated and how to integrate them, aiming to enable a seamless modeling workflow with high usability.

RQ3 How do users perceive the proposed system?

This question evaluates the proposed system from the users' perspective. It aims at understanding how users perceive the system's benefits and limitations. Its results provide insights into the system's strengths and weaknesses and help identify potential design issues and areas for improvement.

Contribution

The proposed and developed novel tool as an artefact of this work that provides a series of functionalities to support process modeling from documents in an innovative way: - It allows users to manage with multiple documents and multiple models for one use case within one unified interface. - It allow a new input method for LLM-based process model generation by allowing users to select multiple passages from one document to serve as input for LLM-based process model auto-generation.

Methodology

This work follows the three interconnected cycles of DSR[1] proposed by Hevner: The relevance cycle: the first cycle is done through the communication with modeling experts, who proposes the tool and eliciate the requirements in section 3.1 for the tool. The rigor cycle: it is done throgh the review in section 2. The design cycle: the first cycle is done through the delivered solution of RQ2 and evaluation result of RQ3. Through the analysis and discussion of the evalution result, we can identify the strengths and weaknesses of the proposed tool, and further improve the design of the tool in the next iteration.

Evaluation

The evaluation of the proposed system is done with 4 experts in qualitative way. It consisted of task-based activities and a follow-up questionnaire to collect user feedback on the tool's usability and perceived usefulness.

Structure

The remainder of this thesis is structured as follows: Chapter 2 reviews the related work of LLM-based process modeling tools. Chapter 3 presents the design solution of the tool, including its requirements, architecture, and functionalities. Chapter 4 describes its technical implementation. Chapter 5 describes the evaluation of the proposed system, including the methodology, results. Chapter 6 discusses the implications of the findings and potential future work. Finally, Chapter 7 summarizes the current work and outlines future work.

Related Work

LLM-based Process Modeling Tool

ProMoAI

ProMoAI [2] is a tool that leverages generative AI to support process modeling. It allows users to create process models by providing natural language descriptions of the processes. The tool uses a large language model (LLM) to generate process models based on the provided descriptions. ProMoAI aims to make process modeling more accessible and efficient by allowing users to create models using natural language, without requiring expertise in process modeling languages or tools.

BPMN-Chatbot

BPMN-Chatbot PMN diagrams, making it more accessible to users who may not be familiar with the technical aspects of BPMN modeling.

AutoBPMN.AI

AutoBPMN.AI [3] is a tool that enables conversational process modeling and automation. It allows users to create and redesign process models through iterative conversations with a large language model (LLM) such as Gemini or GPT. The tool supports both text-based input and existing models, generating new or redesigned process models in executable format. AutoBPMN.AI also features automatic layout capabilities for BPMN-like visualizations. The tool has been demonstrated in various scenarios, including the creation and redesign of process models, as well as subsequent process model execution.

Document-based Interaction Tool

NotebookLM?

NotebookLM **NotebookLM** is a tool that allows users to upload and interact with their documents using a large language model (LLM). It provides a conversational interface for users to ask questions and receive answers based on the content of their documents. NotebookLM is designed to help users quickly find information and gain insights from their documents, making it easier to manage and utilize large amounts of information.

Jupyter Notebook?

Human-in-the-loop

Solution Design

Requirements

Functional Requirements

The system must meet the following core requirements for a MVP:

It shall integrate the necessary modeling functionalities within one unified interface (webpage).

It shall support upload multiple documents in textual format, e.g.,TXT, DOCX, PDF.

It shall allow users to select multiple passages from one document to serve as input for LLM-based process model auto-generation.

It shall provide a clear visual link between the text and the generated model.

It shall allow users to establish process-subprocess relationships between models.

It shall record user behavior and interactions with the system for analysis purposes.

One additional requirement is that the system shall be able to support document update.

Non-functional Requirements

Non-functional requirements for this tool mainly regard technical constraints:

It shall be implemented as a web-based application.

The frontend shall be developed using native web technologies (JavaScript, HTML, and CSS).

The visualization of the process model shall be implemented using the CPEE Layout[1] library.

[1] `https://github.com/etm/cpee-layout`

The source text for the process model should be embedded in the model's XML data.
The LLM-backed process model generation shall be implemented using the API from AutoBPMN.AI.

System Design Overview

The system design overview is described from the user perspective, focusing on how users interact with the system and its core functionalities.

Since the innovation of this tool focuses on the integration of functionalities and an innovative modeling workflow in the frontend, so the system overview is described from the user perspective, since its main value proposition is to provide a seamless and efficient user experience for process modeling based on documents.

The system overview is presented from a user perspective, as the main contribution of the tool lies in the integration of functionalities and the design of a seamless and efficient modeling workflow for process modeling based on documents.

The following concepts are defined: - Project: a use case consisting of one or more documents for which process models are created. - Workspace: the main page where the modeling workflow is performed, from document upload to model export. - Traceability: the ability to navigate between source text and the corresponding process model. - Project graph: a network-style visualization of relationships among models within a project.

The system consists of three main pages: a home page for managing projects, a workspace page where the modeling workflow takes place, and a statistics page for viewing project-related metrics.

Firstly, we define the following concepts to better describe the system overview:
- Project: one use case or one task that the user wants to accomplish, which may involve multiple documents that models needs to be created from.
- Workspace: the unfied page for one project, where the modeling workflow can be accomplish from document upload to model export.
- Traceability: the ability to find the source text on the document for a model and the ability to find the model on the source text from documents.

- Project graph: a network-style visualization of the relationships between different models within a project, which can help users to understand the overall structure and flow of the processes being modeled. Details can be found in the section.

It contains three pages: a home page where a user can create a new project or open an existing project, a workspace page where the modeling workflow happens, and a statistic page where the user can view the statistics of the project.

Modeling Workflow Design

As the core part of the system, this section describes the modeling workflow that happens in the workspace page in Use Case Diagram. Figure: the use case diagram for the interaction of a user within the workspace page.
user –> Upload/Rename/Update/Delete Document
–> View document –> Select Text Passages –> Generate Process Model
–> View Process Model –> Modify Process Model –> Link Process Model with other Models
–> Rename/Delete/Export Process Model

Core Functionalities Design

Document Management

Document Interaction

text selection text selection editing

Process modeling

Human-in-the-loop Process Modeling and Version Control.

Use Case: Model Generation. By click the button an adavance mode is desgin

Use Case: Model Modification. a. Manual edit b. Update selected text and apply the changes. c. Refinement by sending prompt to LLM:
Input: prompt only and the current Model
d. Regeneration:
Input: type a. current model + changed selected text passage b. advanced mode: prompt + current model + changed selected text passage

Use Case: Establishing Process-Subprocess Relationships. - a review step for each LLM-generated model.

Figure 1
Example JSON schema used in the backend for text quotes

```
<textQuoteSelector>
</textQuoteSelector>
```

Augumented Model Data.

Traceability

It is a

Project Graph Visualization

Visualize the

User Interface and Interaction Design

The layout design of the user interface is shown in the figure **??** in a 3-pane layout. The left pane is for document uploading and viewing, the middle pane is for process modeling and the right pane is for process models viewing and project graph visualization. The details are divided as the following: a) Document List b) Document Viewer c) Model Editor c.1) Model Canvas c.2) Model Prompt Editor d) Model List e) Project Graph Visualization. Interacion Design: The workspace support a highly freely interaction/navigation between model and documents.

A model can be triggered to be displayed by the following pathes:

(1). Click on the model in the model list

(2). Click on the text passage that is linked to the model

(3). Click on the model tag displayed above the text passage

(4). Click on the model node in the project graph visualization

A document can be triggered to be displayed by the following pathes:

(1). Click on the document in the document list

(2). Click on the model in the model list, then its corresponding document will be displayed and the linked text passage's position will be auto-anchored.

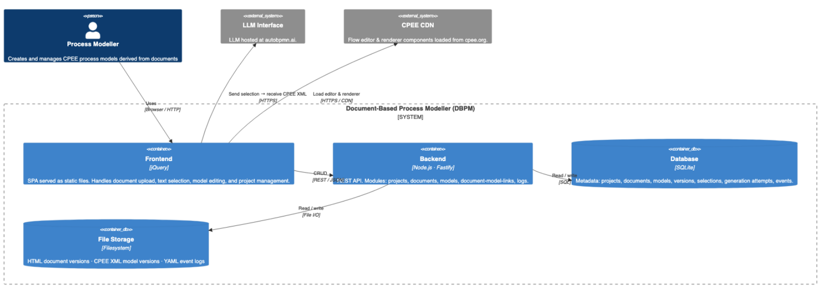
(3). Click on the document node in the project graph visualization

System Architecture

The system architecture is shown in the figure 2. The system is designed as a web-based application, with a front-end implemented using native JavaScript, HTML, and CSS. The back-end is responsible for handling user interactions, managing document uploads, and communicating with the LLM API for process model generation. The visualization of the process models is achieved using the CPEE library, which allows for dynamic rendering of BPMN diagrams based on the generated models.

Figure 2

System Architecture

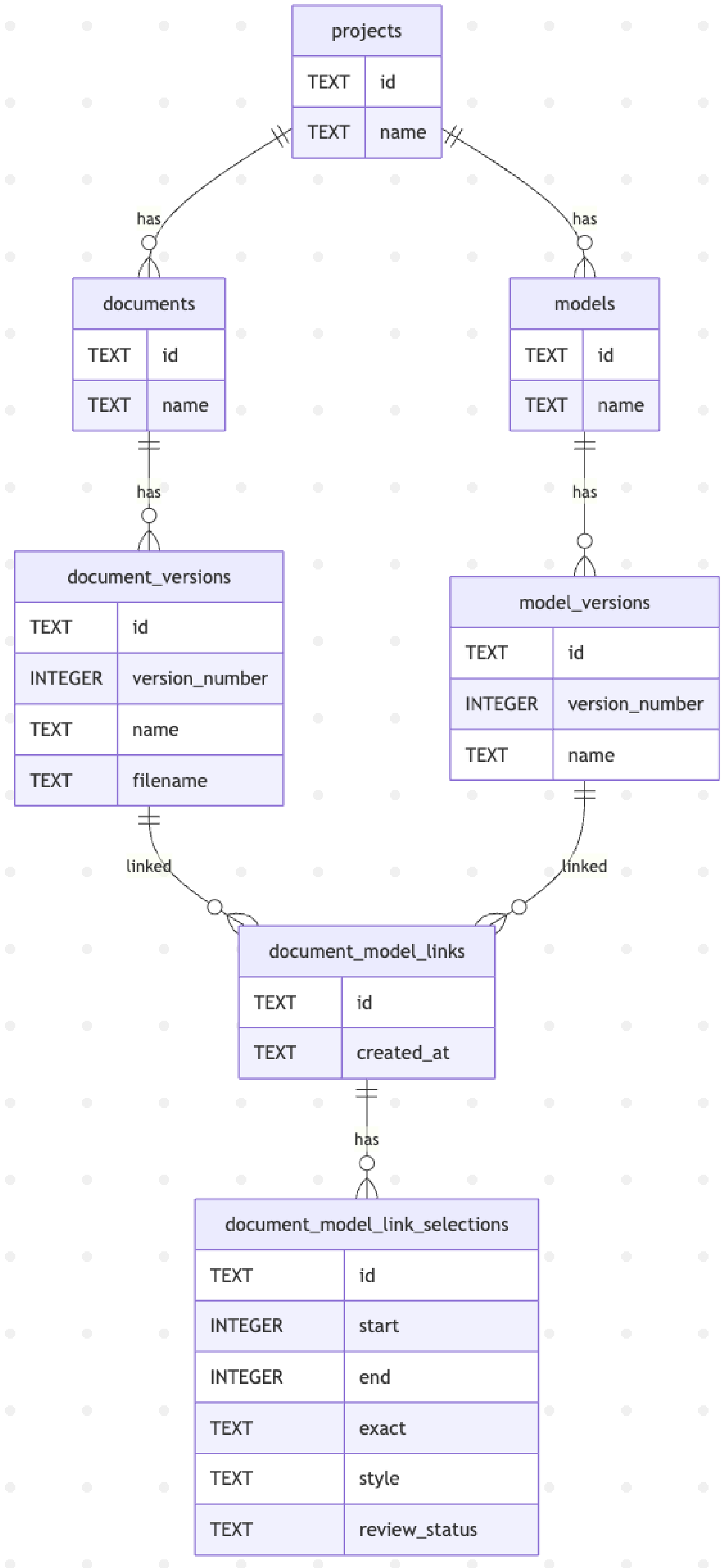


The system architecture of the document-based process modeling tool.

Conceptual Data Model

Figure 3

Conceptual Data Model



The conceptual data model of the document-based process modeling tool.

The system uses a TextQuoteSelector for robust anchoring, as defined in the W3C Web Annotation Model (Text Quote Selector section) [4].

Implementation

Frontend

Architecture

Figure 4

Frontend codebase structure

```
frontend/
├─ api/
│  ├─ base.js
│  ├─ documents.api.js
│  ├─ models.api.js
│  ├─ projects.api.js
│  ├─ document-model-links.api.js
│  └─ ...api.js
├─ modules/
│  ├─ document/
│  │  ├─ file.js
│  │  └─ selection.js
│  └─ model/
│     ├─ wfadaptor.js
│     ├─ endpoints/
│     ├─ themes/
│     └─ utils/
├─ pages/
│  ├─ home/
│  │  ├─ init.js
│  │  └─ ui/
│  │     ├─ projects.ui.js
│  │     ├─ project-creation.ui.js
│  │     └─ dashboard.ui.js
│  ├─ stats/
│  │  ├─ init.js
│  │  ├─ store.js
│  │  └─ ui/
│  └─ workspace/
```

Document Processing

Text Selection Rendering

Virtual Overlay Strategy.

Virtual Overlay Strategy.

Modeling

Rendering Customization

Architecture

Backend

Architecture

The tech-stack for the backend is currently chosen as Node.js with Fastify, as it provides out-of-the-box support for schema validation. As shown in the figure 5, the example code snippet of the backend

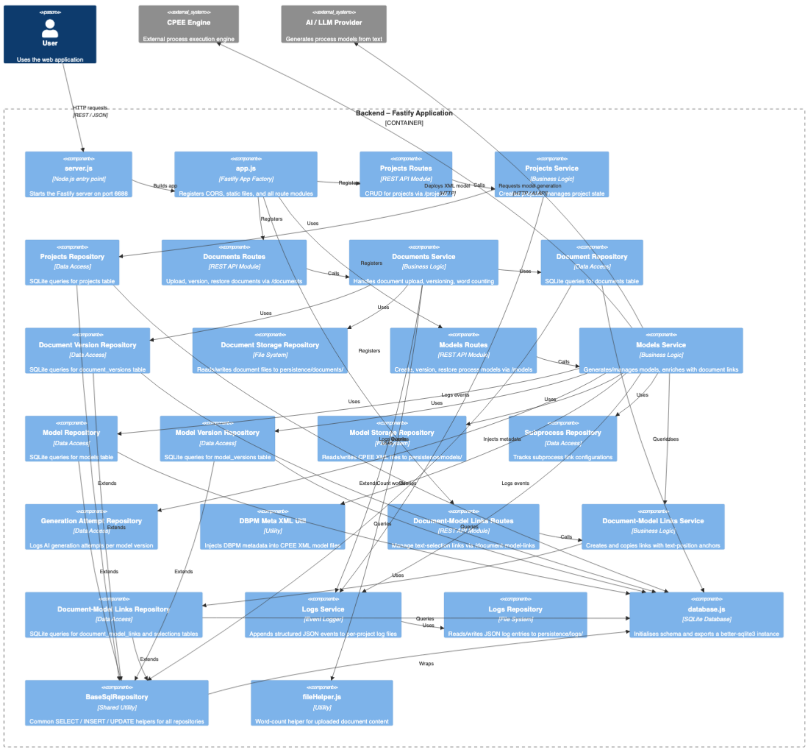
Figure 5
Example JSON schema used in the backend for text quotes

```
const textQuoteSchema = {
  type: "object",
  required: ["exact"],
  additionalProperties: false,
  properties: {
    exact: { type: "string" },
    prefix: { type: "string" },
    suffix: { type: "string" },
  },
};
```

The backend implemention follows the layered Controller-Service-Repository architecture.

Figure 6

Backend Component



The backend component of the document-based process modeling tool.

Database

Evaluation

Evaluation Procedure

The developed system was evaluated with four modeling experts to assess its usability and perceived usefulness, as well as to gather feedback on potential improvements and additional features.

Each evaluation session followed a semi-structured procedure and lasted approximately 40 minutes. It began with a brief demonstration of the system's workflow and main features. After that, the participants were asked to complete two tasks. The first was an exploratory task based on a synthetic B2B order processing scenario consisting of two wiki-style documents in DOCX format (see Appendix A.1). It was designed to help participants familiarize themselves with the system.

The second was a more realistic process modeling task based on an open-source SOP document [2] in PDF format. For both tasks, participants were not given strict instructions on how to proceed, allowing them to explore the system freely. If time permitted, participants could additionally use the system with their own documents.

Following task completion, participants completed a questionnaire including the following questions:
(1) What aspects of the tool did you like while using it?
(2) What aspects of the tool did you not like or find problematic?
(3) What improvements or additional features would you most like to see in the tool?
(4) If you needed to model processes from documents in the future, would you consider using this tool? Why or why not?

Evaluation Results

Overall Perception

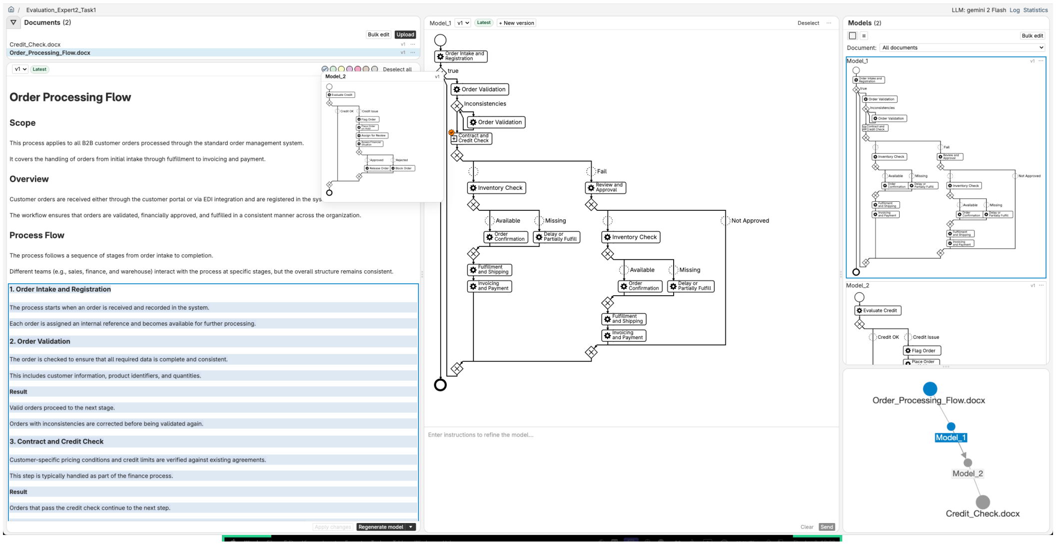
Overall, participants perceived the tool as usable and intuitive, and capable of supporting document-based process modeling tasks within a unified environment. The Figure (See Figure 7) and the figure (See Figure 8) illustrate the modeling results from one expert for the first and second tasks, respectively.

The feedback to each functionality is summarized within the table 1. Due to different background of the expert, different experts provided different perspectives on the system, therefore no quantiative summary is provided in table, but it is mentioned in the description of the following sections when relevant.

[2] DNRC SOP Document

Figure 7

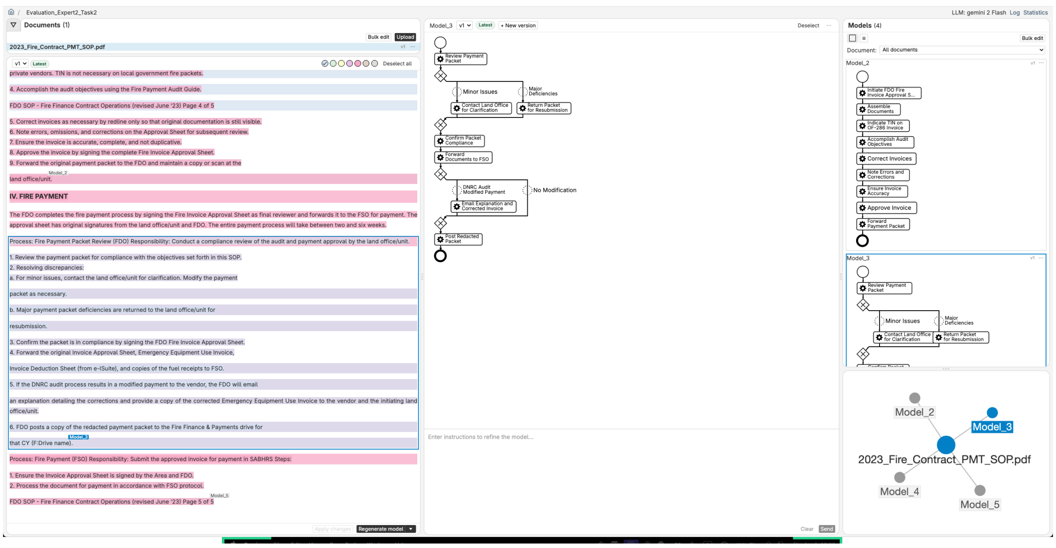
Modeling result



Modeling result of an expert for the first task.

Figure 8

Modeling result



Modeling result of an expert for the second task.

Table 1
Summary of Evaluation Results by Functionality

Functionality	Perceived Benefits		Perceived Limitations	
	Item	n	Item	n
Document Interaction	Intuitive text selection workflow	1	Readability issue	1
Integrated LLM-based Process Modeling	Effective	1	Unnecessary review step of the first generated model	2
	Good quality	1	Lack of transparency in regeneration logic	1
Traceability	Clear	1	Lack of fine-grained link between actual used text and model	2
Process-Subprocess Relationship	Convienient	2		
Model Versioning	Helpful	1		
Project Graph	Helpful for larger systems	1	Node types not clear	1
Log and Statistics	Potentially useful for analysis	1		

n = number of participants who mentioned each item

Positive Aspects

The basic workflow of selecting relevant text, generating process models, and the linkage between the selected text and the corresponding models was generally perceived as clear and easy to follow. In addition, establishing relationships between process models and subprocesses was perceived as simple and useful.

Besides the core functionalities of the system, several additional positive aspects were perceived by participants. First, the versioning functionality was highlighted as the most valuable feature, as it effectively supports iterative process model development by enabling users to track changes and revisit previous model versions. Second, the embed within the model XML data of relevant information such as selected text and prompt history is considered a well-integrated design feature by some participants. Third, logging and statistical information elicited interest among some participants due to its potential for further analysis. Finally, the network-style visualization of relationships across models and documents was mentioned as helpful, especially in the use of larger systems.

Challenges in Interaction and Use

Despite the generally positive perception of the system, participants reported several challenges related to interaction and use. These challenges can be grouped into three categories: interaction-level issues, practical limitations in use, and challenges in understanding the system's underlying logic. At the interaction level, several issues were identified in the system's interaction design. In particular, the requirement to explicitly deselect the current model before initiating a new modeling step was not immediately clear to users. In addition, the review step following the initial generation of models was perceived by some participants as unnecessary, as the generated models were already of sufficiently good quality. At the level of system use, participants reported practical constraints when working with documents. In particular, the display of PDF documents was perceived as suboptimal, which affected their experience when reading. Beyond interaction and practical usage issues, participants also encountered challenges in understanding the system's underlying logic, particularly in relation to AI-assisted model modifications. While the initial generation of models from selected text was generally well understood, subsequent AI-assisted modifications were less transparent to users. It was often unclear how the system constructed its input, including whether and how the current model and selected text were incorporated into the generation process. As a result, participants found it difficult to anticipate how their actions would influence the generated outputs, which occasionally led to mismatches between expected and actual results.

Potential Improvements and Additional Features

Beyond the challenges identified in system use and interaction, a number of improvements and desired features for the system were suggested. These can be grouped into two categories: system-level features and more advanced AI-assisted modeling capabilities. At the system level, improvements and desired features were suggested across several aspects Firstly, improvements to the visualization were mentioned, including clearer differentiation and identification of elements (i.e. which type of element a node is and which version it represents). More expressive visual representations are also suggested such as using size, distance, or other visual properties to convey additional information. Secondly, model comparison was suggested as a valuable feature to help users understand and verify changes between different model versions, making it easier to identify modifications and track model iterations over time. Finally, an AI-assisted document interaction feature was suggested, where relevant information can be retrieved from documents and automatically selected to reduce manual effort and time. Some minor UI improvement suggestions are not described here in details. The second category concerns advanced AI-assisted modeling capabilities. In particular, participants emphasized the need for fine-grained traceability between source text and model. When selecting larger text passages, only a subset of the text is reflected in the generated model, and it is unclear which parts are actually used. Therefore, a clearer indication of the actual text used for model generation is required, ideally at the level of individual tasks. In addition, a retrieval-augmented generation (RAG)-based approach was proposed, in which all documents can serve as context for model generation. Beyond this, participants also expected more general advanced AI modeling capabilities, including explainability of the generated model through conversational interaction with chatbots, as well as support for model quality evaluation and automatic improvement.

Future Use Intention

Participants indicated an intention to use the system in use cases involving the management and exploration of multiple models and their associated documents. In these cases, features such as clear text-model traceability, process-subprocess relationship, and model management functionalities were considered particularly useful. In contrast, for simpler use cases, such as generating a process model from a single document, the system remains a viable option but does not provide a strong advantage over existing approaches. In general, participants pointed to the need for more advanced functionalities, specifically in relation to AI-assisted modeling features and improved interaction with documents.

Additional Observations

Besides the gathered feedback, some findings regarding user behaviour and interaction are observed during the evaluation.

While most participants tended to select larger portions of text when starting with a new document, some preferred to begin with building models for smaller segments. These different strategies are illustrated in Figure 9.

Figure 9

Different modeling strategies

Different modeling strategies observed during task execution (above: larger part selection; below: step-by-step selection)

Discussion

This chapter discusses the findings of the evaluation and reflects on their implications for system design.

Usability and User Experience

There're some friction reported during the evaluation. While some features can be mitigated by design, some are hard. A this tool eliciate a new way of process modeling, it requires a little learnability to let user get used to the new way of modeling.

User Expectations

Conclusion

Summary

In this work, we propose and develop a process modeling tool, DBPM, with a series of convienient functionalities, especially for the case of process modeling from documents.

It provides a new modeling workflow for process modeling from documents, which allows users to upload documents, selects passages from the documents and auto-generate process models. The tool also maintains a clear link between the text and the generated model, and allows users to establish process-subprocess relationships between processes and subprocesses. We evaluate the tool with 4 experts from both academia and industry, and the results show that the tool is usable and intuitive, and capable of supporting document-based process modeling tasks within a unified environment. It provides a new interactive approach between the source text and the its corresponding model.

Future Work

Future work includes: - Improving document rendering - Fine-tuning the LLM for more precise text extraction

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Appendix

Evaluation Material

B2B Order Processing Scenario

This use case consists of two synthetic documents.

Document 1: Order Processing Flow (excerpt)

Scope

This process applies to all B2B customer orders processed through the standard order management system. It covers the handling of orders from initial intake through fulfillment to invoicing and payment.

Overview

Customer orders are received either through the customer portal or via EDI integration and are registered in the system before further processing. The workflow ensures that orders are validated, financially approved, and fulfilled in a consistent manner across the organization. Process Flow The process follows a sequence of stages from order intake to completion. Different teams (e.g., sales, finance, and warehouse) interact with the process at specific stages, but the overall structure remains consistent. 1. Order Intake and Registration The process starts when an order is received and recorded in the system. Each order is assigned an internal reference and becomes available for further processing. 2. Order Validation The order is checked to ensure that all required data is complete and consistent. This includes customer information, product identifiers, and quantities. Result Valid orders proceed to the next stage. Orders with inconsistencies are corrected before being validated again. 3. Contract and Credit Check Customer-specific pricing conditions and credit limits are verified against existing agreements. This step is typically handled as part of the finance process. Result Orders that pass the credit check continue to the next step. Orders that fail the check are placed on hold for review and require approval before processing can continue. 4. Inventory Check The system verifies whether the requested items are available in stock. Result Orders with available items proceed to confirmation. Orders with missing items are delayed or partially fulfilled, depending on stock availability. 5. Order Confirmation Once the required checks are completed, the order is confirmed and released for execution. At this stage, the order is considered ready for fulfillment. 6. Fulfillment and Shipping The warehouse prepares the order for shipment, including picking and packing. The goods are then shipped by the logistics provider. 7. Invoicing and Payment An invoice is generated and sent to the customer. Payment is processed according to the agreed terms. The process is considered complete once the invoice has been issued and payment is recorded in the system. Notes This document provides a high-level view of the order processing flow. Detailed rules and system-specific behavior are described in the corresponding process pages. Orders that are placed on hold remain in the process until the required issues have been resolved and approval has been granted. Related Pages For more detailed rules and step-specific behavior, see: - Order Validation - Credit Check

Document 2: Credit Check (excerpt) The credit check is performed after order validation and includes automated evaluation, issue detection, manual review, and decision-making.

Something Text here.

Next paragraph.